

Smart traffic signals help blind cross streets

Smart traffic signals that are designed to improve the traffic flow could also help pedestrians with visual or other disabilities safely cross streets, or even catch a bus.

Researchers at Carnegie Mellon University's Robotics

Institute have begun a two-year project to develop a system that relays information from a user's smartphone directly to the smart traffic signals. The signals could adjust their timing to accommodate users who need extra time to cross.

"The smartphone can learn how fast the pedestrian moves, or if the user might have difficulty at certain intersections," explained Stephen Smith, professor of robotics. "The intersection could extend the green in real time to give her the extra time she needs. And it might monitor the phone's location so that it notices if she starts moving outside of the crosswalk."

Smith had previously led development of the Surtrac system of smart signals. The signals use artificial intelligence to monitor traffic; these can alter their timing to maximise the number of cars that can pass through an intersection and keep cars from idling needlessly at a stop light when no cross traffic is present.

Development of the system for people with disabilities will initially involve a set of existing smart signals. The smartphones will need to be equipped with dedicated short-range communications (DSRC) radios to communicate with the signals.



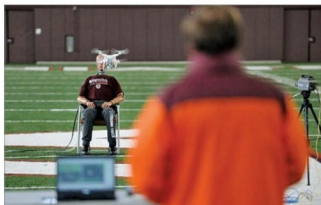
Smart traffic signals (Image courtesy: www.cmu.edu/)

Study weighs risks of drone-human collisions

If drones are to take on tasks like package delivery and traffic management, these will have to fly over humans, raising the risk of drone-human collisions. And, these risks vary widely, researchers at Virginia Polytechnic Institute and State University have found in a comprehensive survey.

Using a crash test dummy, Virginia Tech researchers analysed the potential risk of head and neck injuries posed by three commercially available drones in various impact scenarios. The dummy was outfitted with sensors to measure acceleration and force during the different impacts.

Researchers flew the three aircrafts at full speed into the dummy's head at different angles. They also dropped the aircrafts onto the dummy's head at various orientations. Acceleration and force measurements were used to estimate the risk of severe or life-threatening injuries, like a skull fracture. The smallest and lightest of the three drones,



To measure the risks of commercial drone use, researchers flew several models into the head of a crash-test dummy (Image courtesy: Virginia Tech Mid-Atlantic Aviation Partnership/Institute for Critical Technology and Applied Science)